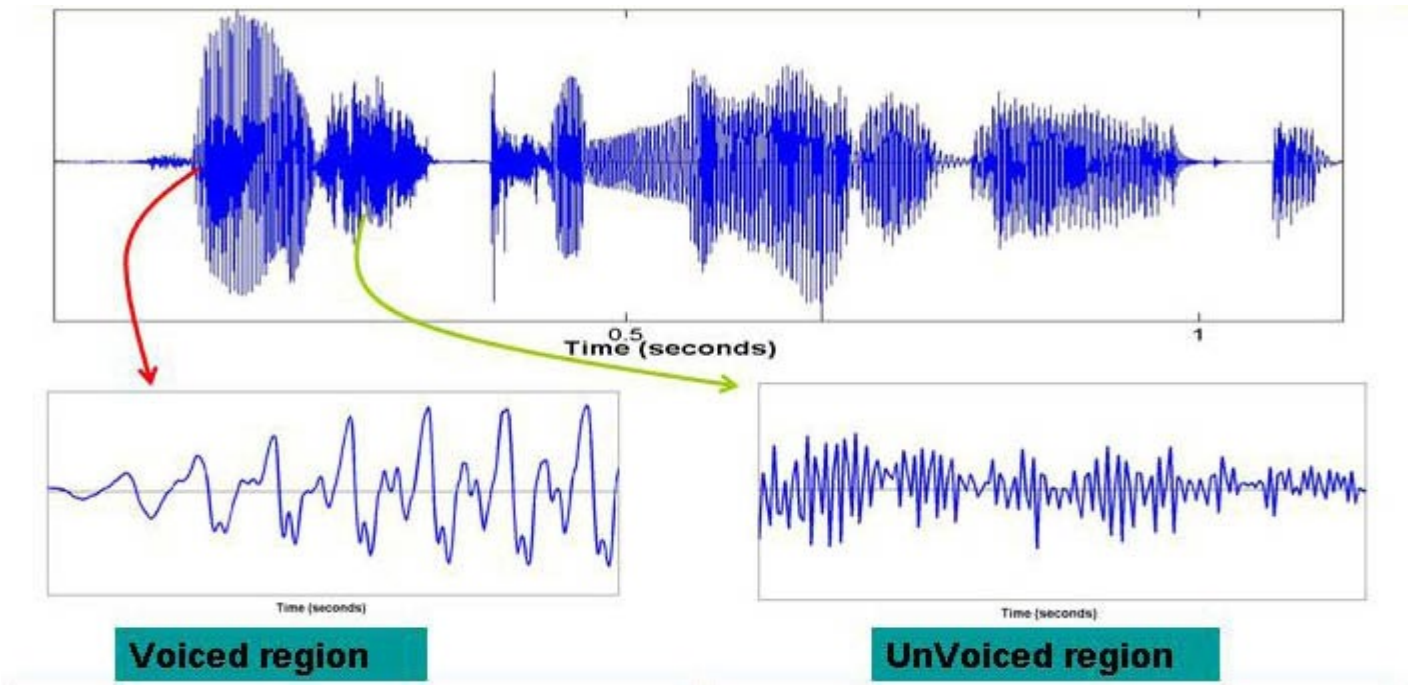
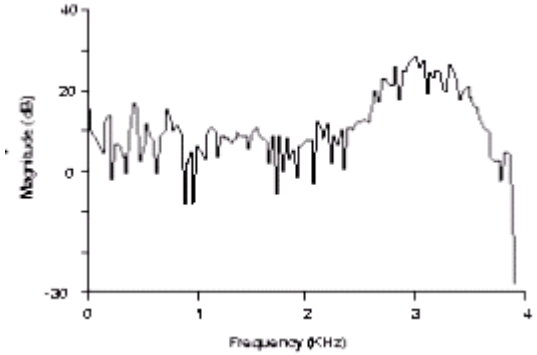
Voiced/Unvoiced Speech Characteristics



Plosive: Pizza,pepper,top
Fricatives sound: when teeth touch the bottom lips e.g. Fire,laughter



Question 1

Briefly explain within the context of Speech Recognition (SR) the following term:

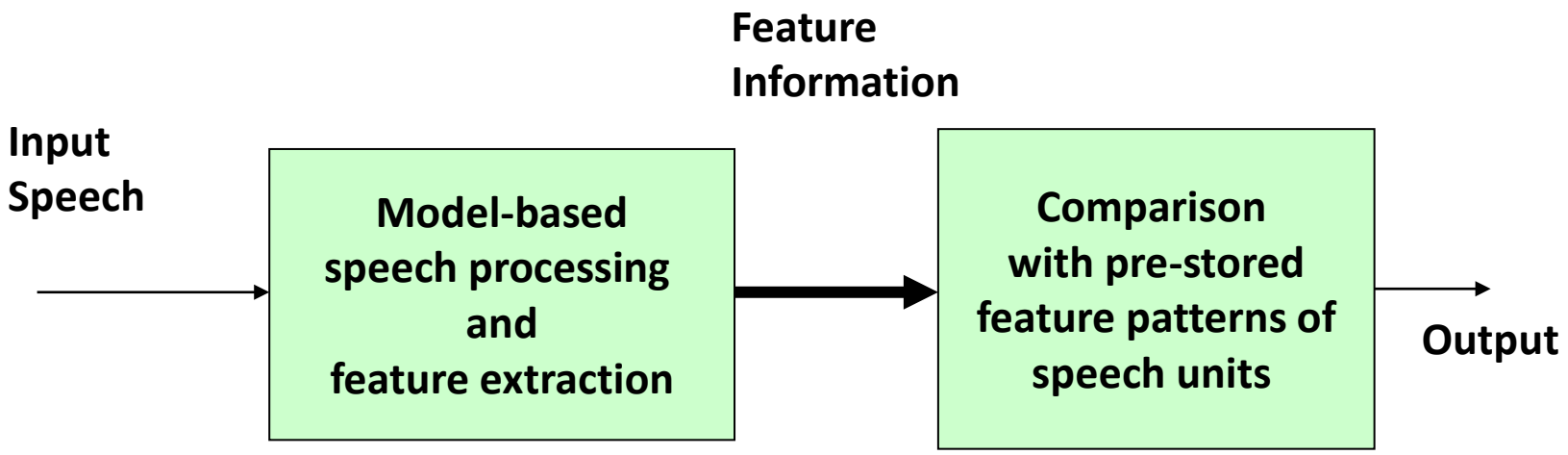
Speaker Dependent SR

Speaker Independent SR systems

SD-SR uses the sounds of a **single** speaker to learn the parameters or models that characterise the system's internal model of the speech process. Thus the system is used to recognize the speech of its trainer.

SI-SR systems are trained by **multiple** speakers and used to recognize even speakers outside the training population.

For the same level of complexity, **SD** systems are generally more accurate than **SI** systems.


Feature
Information
Input
Speech
Model-based
speech processing
and
feature extraction
Comparison
with pre-stored
feature patterns of
speech units
Output

- Speech Recognition system must be capable of:
  - Recognizing continuous speech from multiple speakers with
    - diverse accents,
    - speaking styles,
    - different vocabularies and
    - grammatical tendencies, even multiple languages in noisy environments.
  - Adapting and learning new information as a human can.

Question 2

Briefly explain the following terms:

Speaker identification

Speaker verification

Identification and Verification

Can be:

- Text-Dependent
- Text-Independent

Question 3

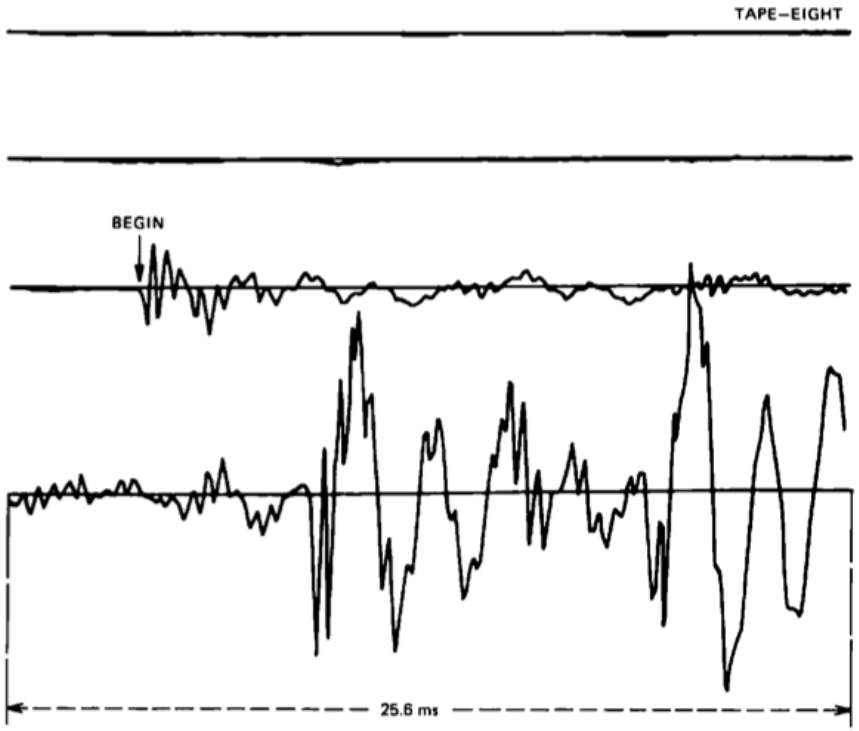
Briefly describe the role of Endpoint detection in a SR system.

Endpoint Detection

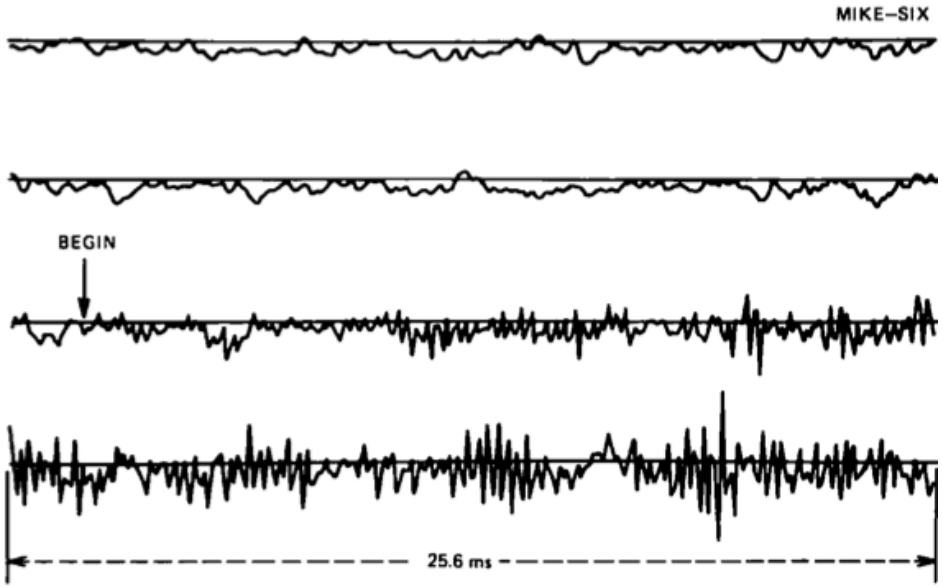
- Detect the presence of speech in a background of noise. Also, accurately detect the beginning and the end of an utterance. 在噪声背景下检测语音的存在
- The task of separating speech from background "silence" is **trivial** only in acoustic environments with extremely **high signal-to-noise ratios** (SNRs).
  - ✓ The energy of the lowest-level speech sounds will exceed the background noise energy.
  - ✓ Hence, a simple energy measure suffices.
- The following figures show the impact of different recording environments on endpoint detection.
  - ✓ TAPE: High-quality recording on tape in soundproof room.
  - ✓ MIC: Recording on computer memory using noise-reducing microphone.

Endpoint Detection

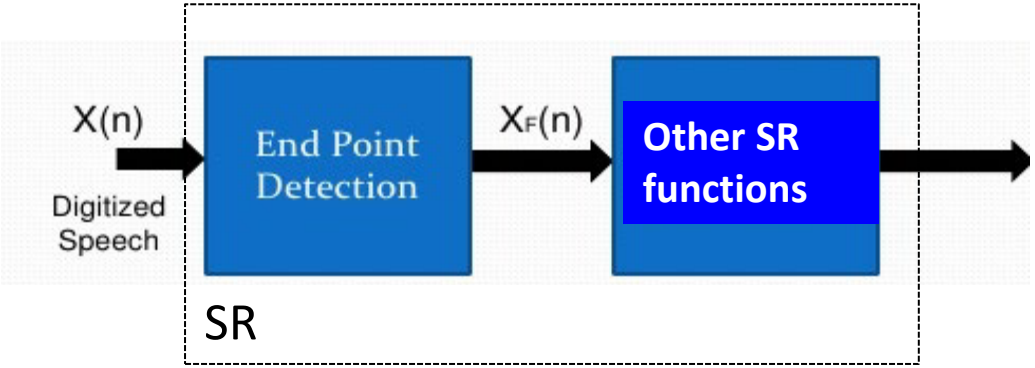
TAPE: The silence pattern is easily distinguished from the speech (word ‘eight’). Observe the radical change in waveform **energy** between silence and the beginning of speech.



MIC: The frequency content of the speech (word ‘six’) is radically different from the frequency content of the background noise, as demonstrated by the sharp increase in **zero-crossing rate** of the waveform.



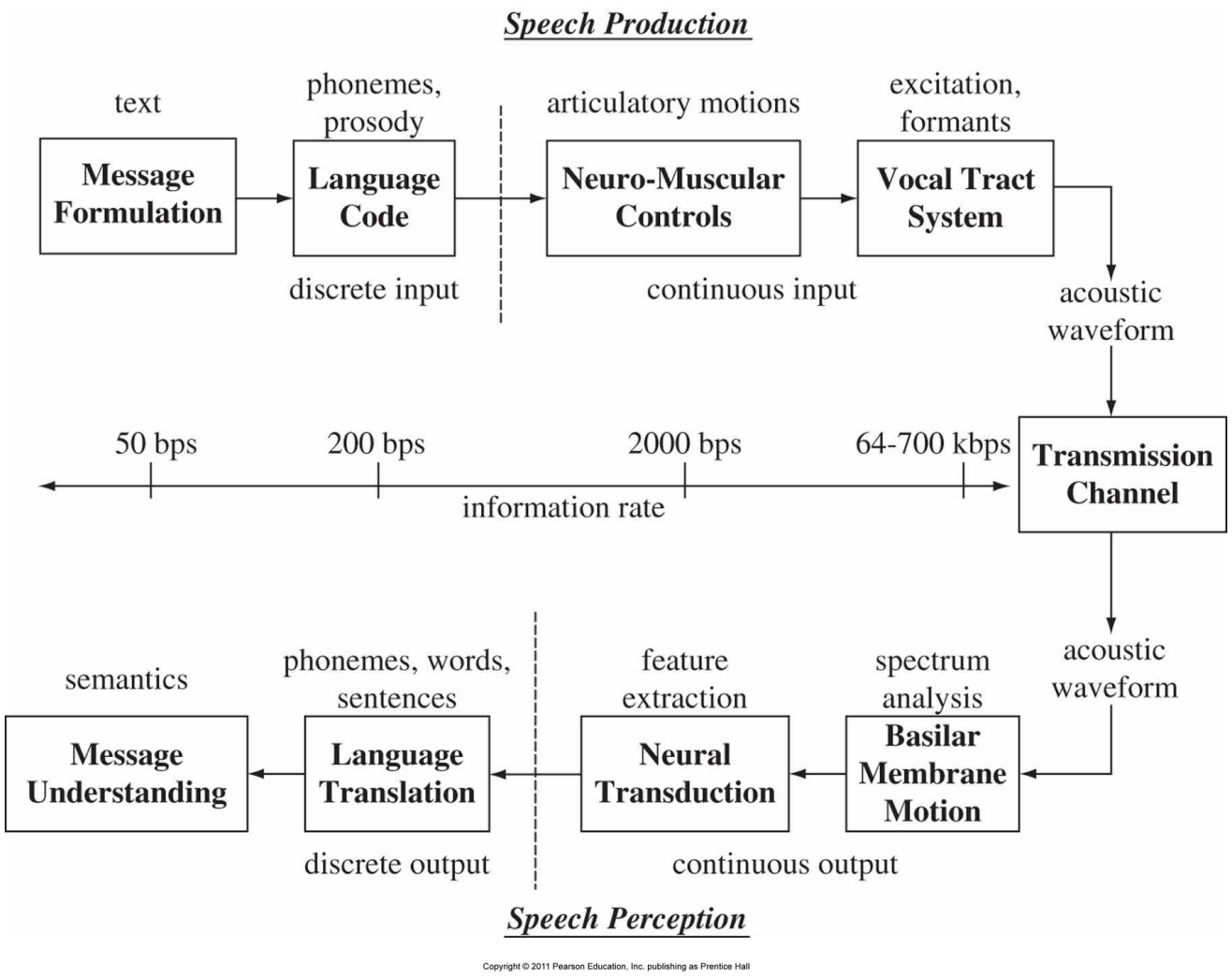
- Endpoint detection provides initial estimates or sets of estimates for use in higher SR system stages.
- Also important for setting the *end-of-speech timeout*, i.e., the length of the pause before the system decides that the user has finished speaking.



Question **4**

Provide a functional block diagram of a Generic Speech Production and Perception system.

Process of producing and perceiving speech



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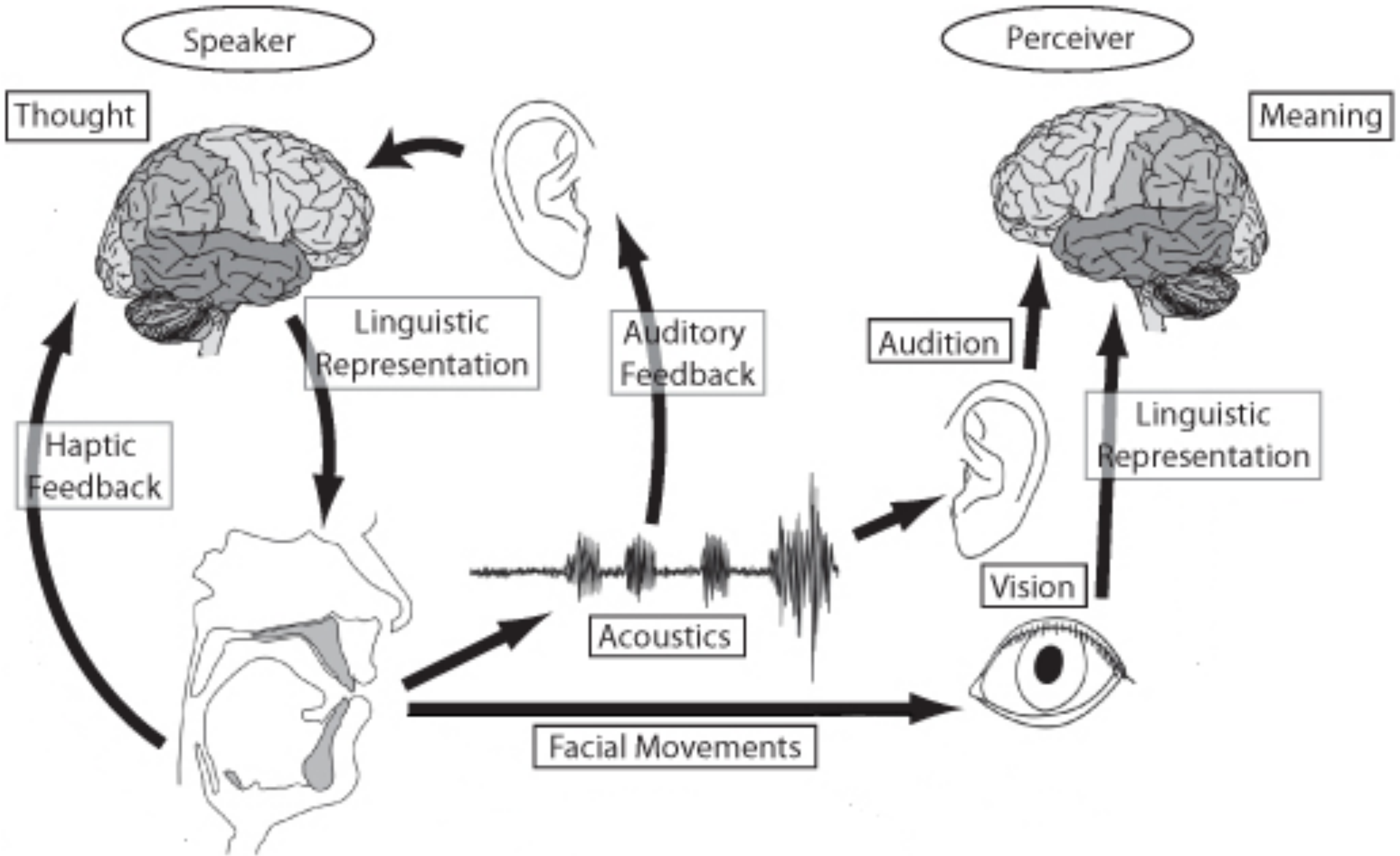
The Speech Chain: from message, to speech signal, to understanding.


Speaker
Perceiver
Thought
Meaning
Linguistic Representation
Auditory Feedback
Audition
Linguistic Representation
Haptic Feedback
Vision
Acoustics
Facial Movements

Question 5

The level of difficulty in achieving high speech recognition performance and system complexity

1. Increases exponentially with vocabulary size
2. Increases logarithmically with vocabulary size
3. Increases linearly with vocabulary size
4. Remains constant with vocabulary size
5. None of the above

- No new Lab this week